

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 17

Sense Organs: The Eye and Ear

Chapter Goals (Slide 1 of 2)

- Identify locations and functions of the major parts of the eye and ear.
- Name the combining forms, prefixes, and suffixes most commonly used to describe these organs and their parts.
- Describe the abnormal conditions that may affect the eye and ear.

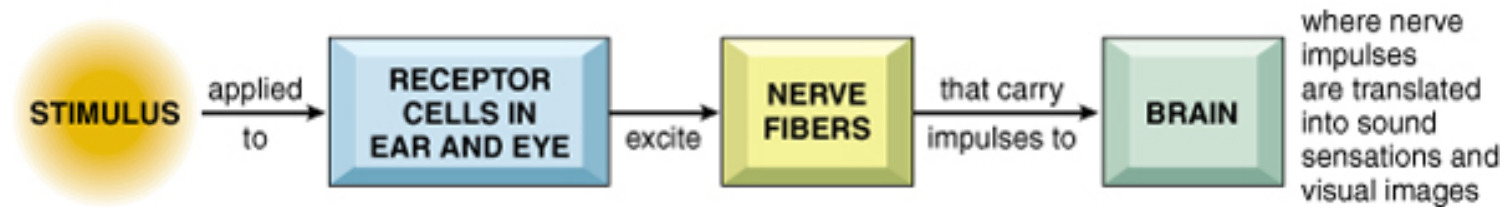
Chapter Goals (Slide 2 of 2)

- Identify clinical procedures that pertain to ophthalmology and otology.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

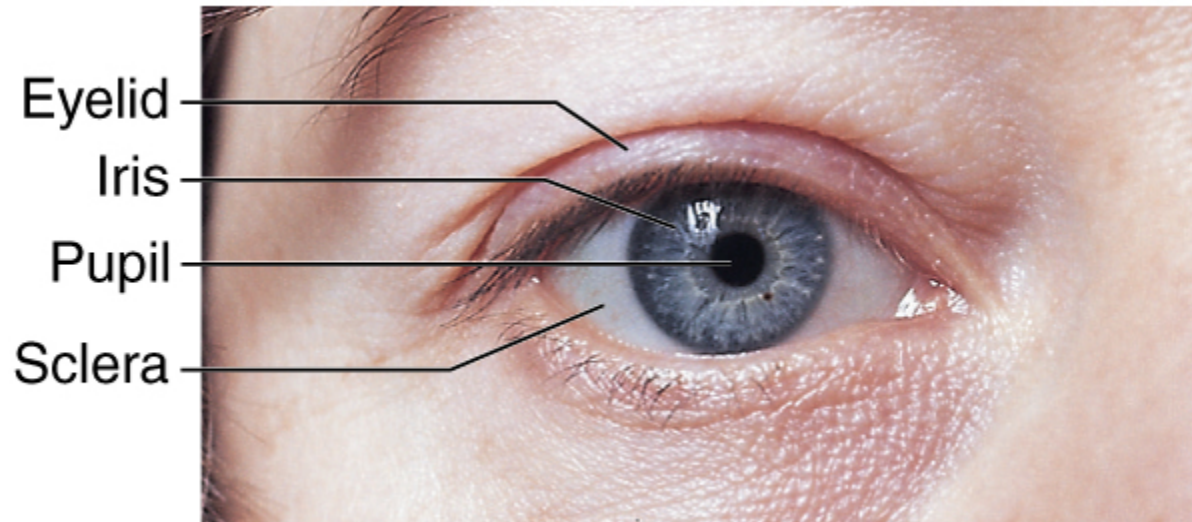
Chapter 17

Lesson 17.1

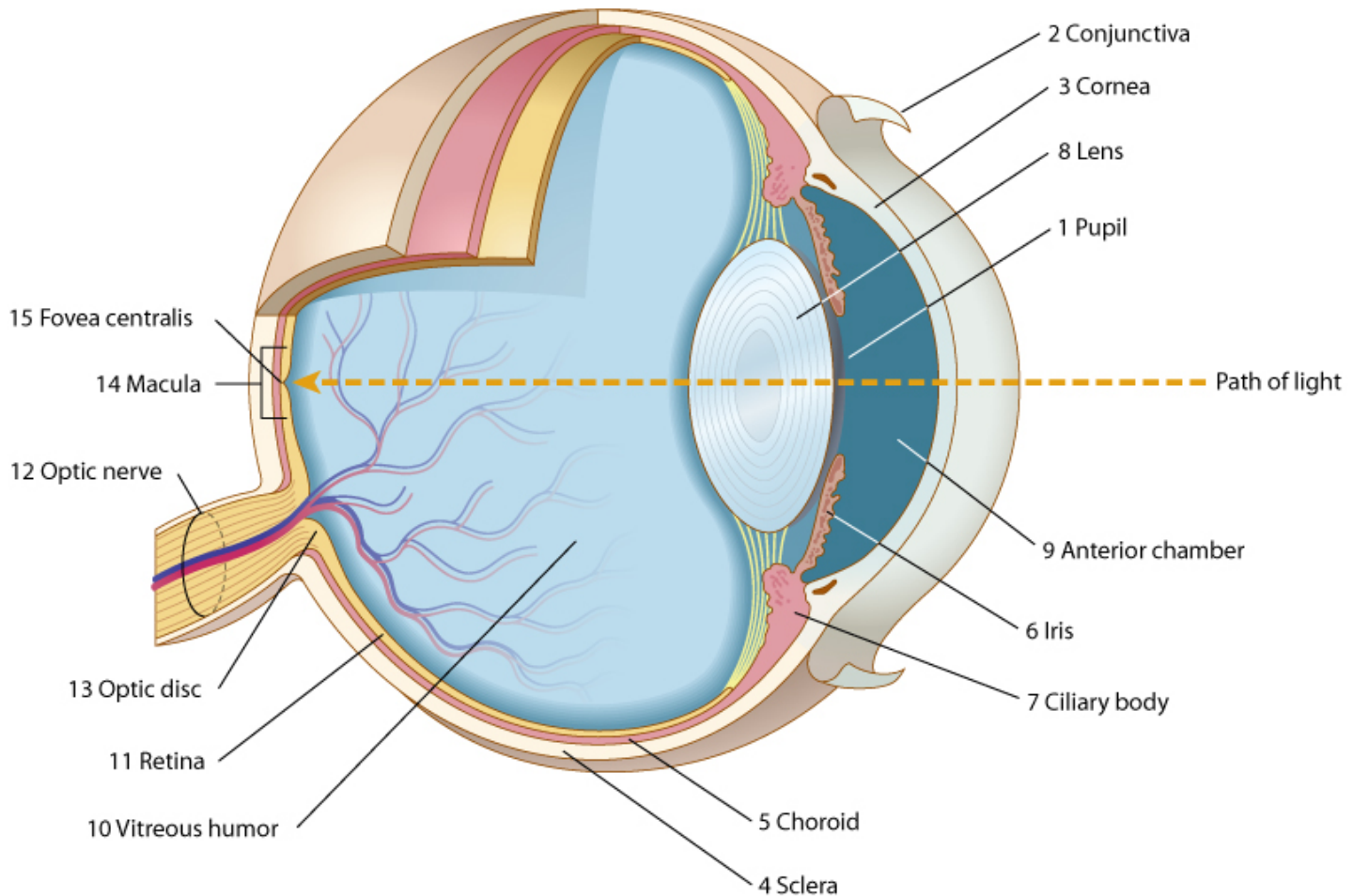
Introduction



The Eye (Slide 1 of 3)



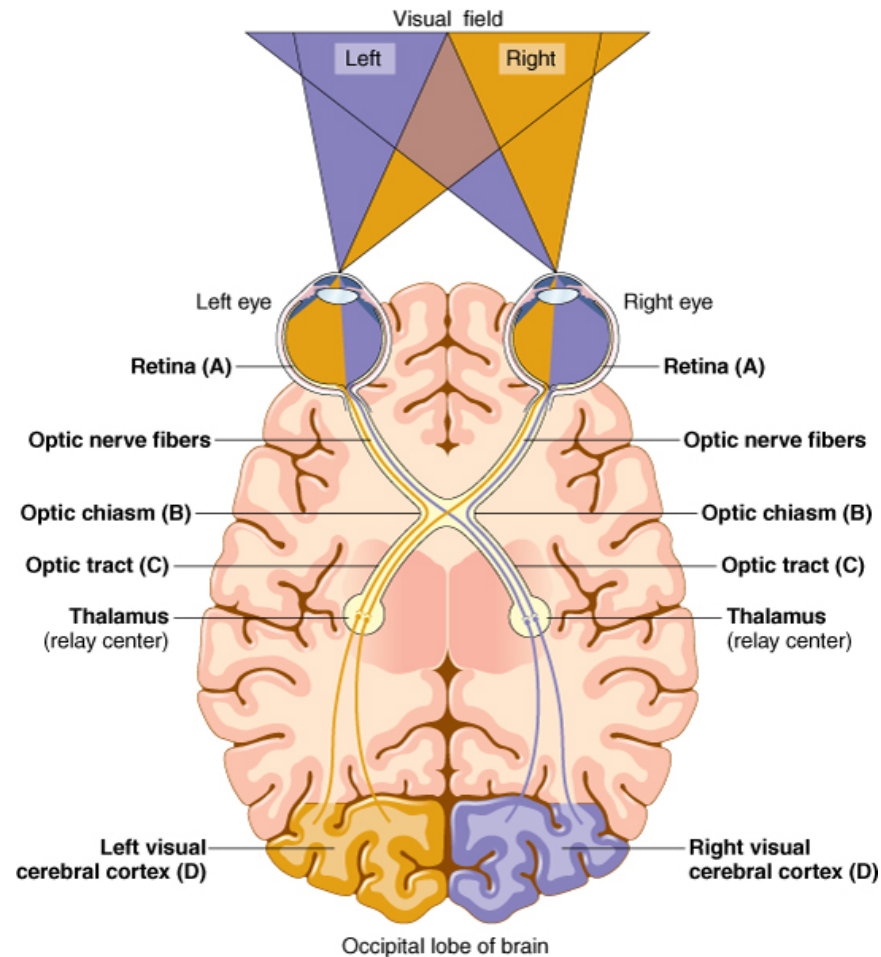
The Eye (Slide 2 of 3)



The Eye (Slide 3 of 3)

- Pupil
- Conjunctiva
- Cornea
- Sclera
- Choroid
- Iris
- Ciliary body
- Lens
- Fundus
- Anterior chamber
- Aqueous humor
- Vitreous chamber
- Vitreous humor
- Retina
- Optic nerve
- Optic disc
- Macula
- Fovea centralis

Visual Pathways from Retina to Cortex



QUICK QUIZ (Slide 1 of 3)

- 1. What is the soft, jellylike material behind the lens in the vitreous chamber that helps maintain the shape of the eyeball?**
 - A. Sclera
 - B. Vitreous humor
 - C. Aqueous humor
 - D. Fovea centralis

Vocabulary – The Eye (Slide 1 of 16)

Term
accommodation
anterior chamber
aqueous humor

Vocabulary – The Eye (Slide 2 of 16)

Term	Meaning/Definition
accommodation	Normal adjustment of the eye to focus on objects from far to near
anterior chamber	The area behind the cornea and in front of the lens and iris
aqueous humor	Fluid produced by the ciliary body and found in anterior chamber

Vocabulary – The Eye (Slide 3 of 16)

Term
biconvex
choroid
ciliary body

Vocabulary – The Eye (Slide 4 of 16)

Term	Meaning/Definition
biconvex	Consisting of two surfaces that are rounded, elevated, and curved evenly; lens of the eye is biconvex
choroid	Middle, vascular layer of the eye, between the retina and the sclera
ciliary body	Structure surrounding the lens that connects the choroid and iris; controls shape of lens and secretes aqueous humor

Vocabulary – The Eye (Slide 5 of 16)

Term
cone
conjunctiva
cornea

Vocabulary – The Eye (Slide 6 of 16)

Term	Meaning/Definition
cone	Photoreceptor cell in the retina that transforms light energy into a nerve impulse; responsible for color and central vision
conjunctiva	Delicate membrane that lines the undersurface of eyelids and covers anterior eyeball
cornea	Fibrous transparent layer of clear tissue that extends over the anterior portion of

Vocabulary – The Eye (Slide 7 of 16)

Term
fovea centralis
fundus of the eye
iris

Vocabulary – The Eye (Slide 8 of 16)

Term	Meaning/Definition
fovea centralis	Tiny depression in the retina that is the region of clearest vision
fundus of the eye	Posterior, inner part of the eye
iris	Pigmented layer that opens and closes to allow more or less light into the eye

Vocabulary – The Eye (Slide 9 of 16)

Term
lens
macula
optic chiasm

Vocabulary – The Eye (Slide 10 of 16)

Term	Meaning/Definition
lens	Transparent, biconvex body behind the pupil of the eye
macula	Yellowish region on the retina lateral to and slightly below the optic disc
optic chiasm	The point at which optic nerve fibers cross in the brain

Vocabulary – The Eye (Slide 11 of 16)

Term
optic disc
optic nerve
pupil

Vocabulary – The Eye (Slide 12 of 16)

Term	Meaning/Definition
optic disc	The region at the back of the eye where the optic nerve meets the retina; blind spot
optic nerve	Cranial nerve carrying impulses from the retina to the brain
pupil	Central opening of the eye, surrounded by the iris, through which light rays pass

Vocabulary – The Eye (Slide 13 of 16)

Term
refraction
retina
rod

Vocabulary – The Eye (Slide 14 of 16)

Term	Meaning/Definition
refraction	Bending of light rays by the cornea, lens, and fluids of the eye to bring the rays into focus on the retina
retina	Light-sensitive nerve cell layer of the eye containing photoreceptor cells
rod	Photoreceptor cell of retina essential for vision in dim light and for peripheral vision

Vocabulary – The Eye (Slide 15 of 16)

Term
sclera
thalamus
vitreous humor

Vocabulary – The Eye (Slide 16 of 16)

Term	Meaning/Definition
sclera	Tough, white outer coat of the eyeball
thalamus	Relay center of the brain
vitreous humor	Soft, jelly-like material behind the lens in the vitreous chamber

Terminology – Structures and Fluids (Slide 1 of 4)

Combining Form	Meaning
aque/o	water
blephar/o	eyelid
conjunctiv/o	conjunctiva
cor/o	pupil
corne/o	cornea
cycl/o	ciliary body

Terminology – Structures and Fluids (Slide 2 of 4)

Combining Form	Meaning
dacry/o	tears, tear duct
ir/o	iris
irid/o	iris
kerat/o	cornea
lacrim/o	tears
ocul/o	eye

Terminology – Structures and Fluids (Slide 3 of 4)

Combining Form	Meaning
ophthalm/o	eye
opt/o	eye, vision
optic/o	eye, vision
palpebr/o	eyelid
papill/o	optic disc
phac/o	lens of the eye

Terminology – Structures and Fluids (Slide 4 of 4)

Combining Form	Meaning
phak/o	lens of the eye
pupill/o	pupil
retin/o	retina
scler/o	sclera (white of the eye)
uve/o	uvea
vitre/o	glassy

Terminology – Conditions (Slide 1 of 2)

Combining Form	Meaning
ambly/o	dull, dim
dipl/o	double
glauc/o	gray
mi/o	smaller, less
mydr/o	widen, enlarge
nyct/o	night

Terminology – Conditions (Slide 2 of 2)

Combining Form	Meaning
phot/o	light
presby/o	old age
scot/o	darkness
xer/o	dry

Terminology – Suffixes--Conditions

Suffix	Meaning
-opia	vision
-opsia	vision
-tropia	to turn

QUICK QUIZ (Slide 2 of 3)

2. Which term means inflammation of the eyelid?

A. Ophthalmoplegia

B. Keratitis

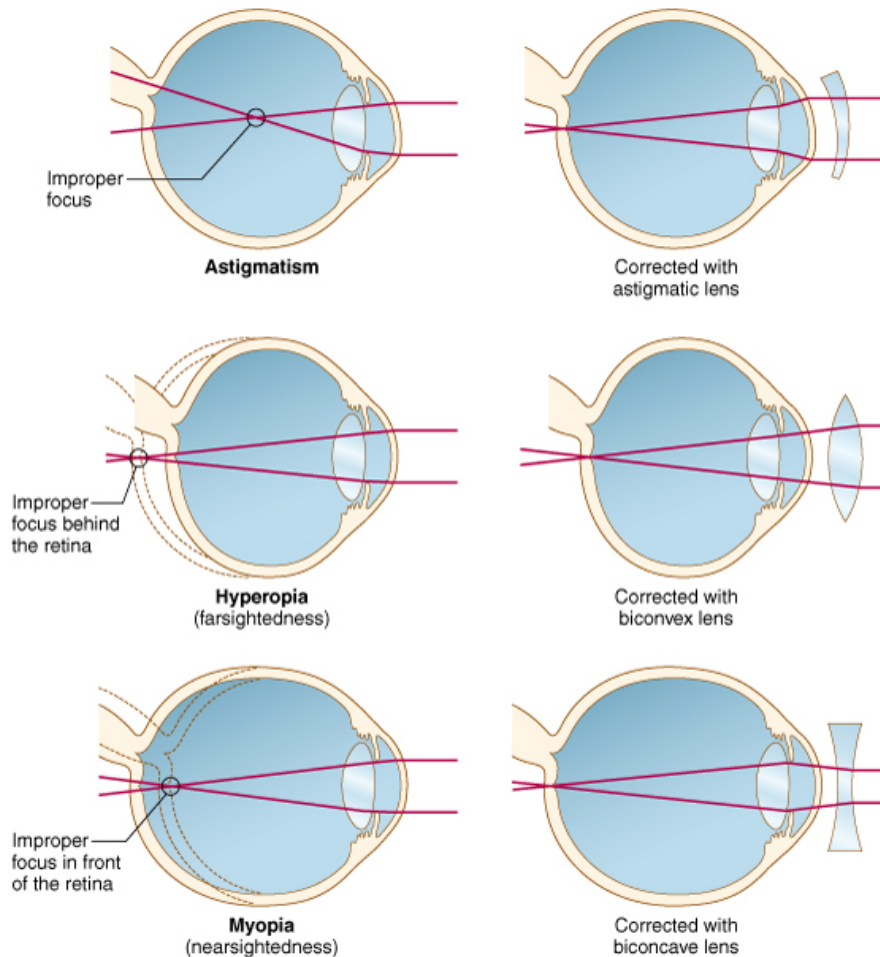
C. Blepharitis

D. Blepharoptosis

Errors of Refraction (Slide 1 of 2)

- **Astigmatism:** defective curvature of the cornea or lens of the eye
- **Hyperopia:** farsightedness
- **Myopia:** nearsightedness
- **Presbyopia:** impairment of vision due to old age reducing lens accommodation

Errors of Refraction (Slide 2 of 2)



Abnormal Conditions (Slide 1 of 6)

Cataract: Clouding of the Lens



Abnormal Conditions (Slide 2 of 6)

Chalazion: a small, hard, cystic mass on the eyelid; formed as a result of chronic inflammation of sebaceous gland along margin of eyelid



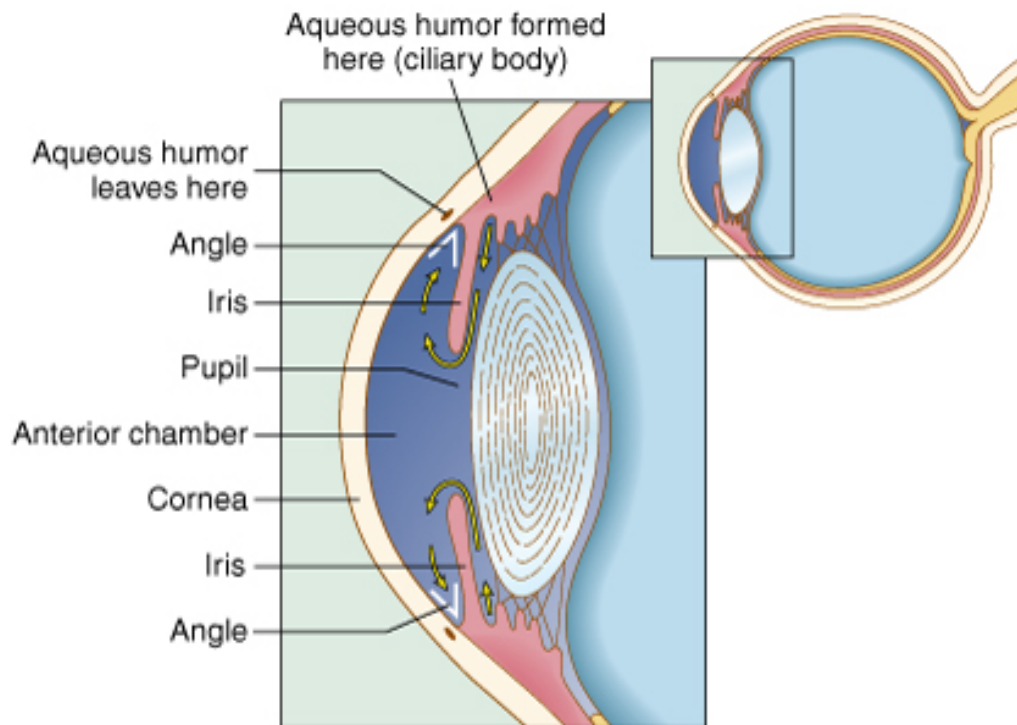
Abnormal Conditions (Slide 3 of 6)

- **Diabetic retinopathy:** retinal effects caused by diabetes mellitus include:
 - Microaneurysms
 - Hemorrhages
 - Dilation of retinal veins
 - Neovascularization

Abnormal Conditions

(Slide 4 of 6)

Glaucoma: increased intraocular pressure damages the retina and optic nerve



Abnormal Conditions (Slide 5 of 6)

- **Hordeolum (stye):** staph infection of a sebaceous gland in the eyelid
- **Macular degeneration:** progressive damage to the macular of the retina
- **Nystagmus:** repetitive rhythmic movements of one or both eyes
- **Strabismus:** abnormal deviation of the eye

Abnormal Conditions (Slide 6 of 6)

- **Retinal detachment:** two layers of the retina separate from each other
 - **Photopsia:** bright flashes of light
 - **Floaters:** vitreous clumps of retina
 - **Scleral buckle:** a belt used to buckle the retina to the sclera
 - **Pneumatic retinopexy:** a gas bubble is injected in vitreous cavity to help reattach the retina

Chapter 17

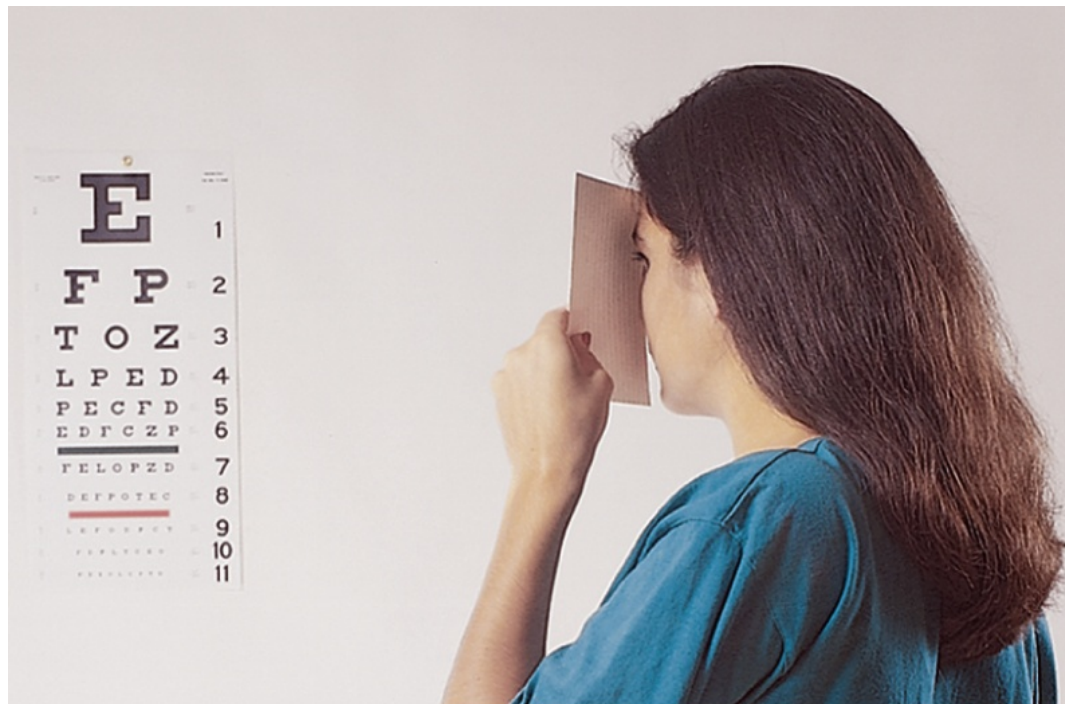
Lesson 17.2

Diagnostic Procedures (Slide 1 of 3)

- **Fluorescein angiography:** dye injection to examine blood flow in the retina
- **Ophthalmoscopy:** visual examination of the interior of the eye through dilated pupil
- **Slit lamp microscopy:** magnified view of expanded number of eye structures

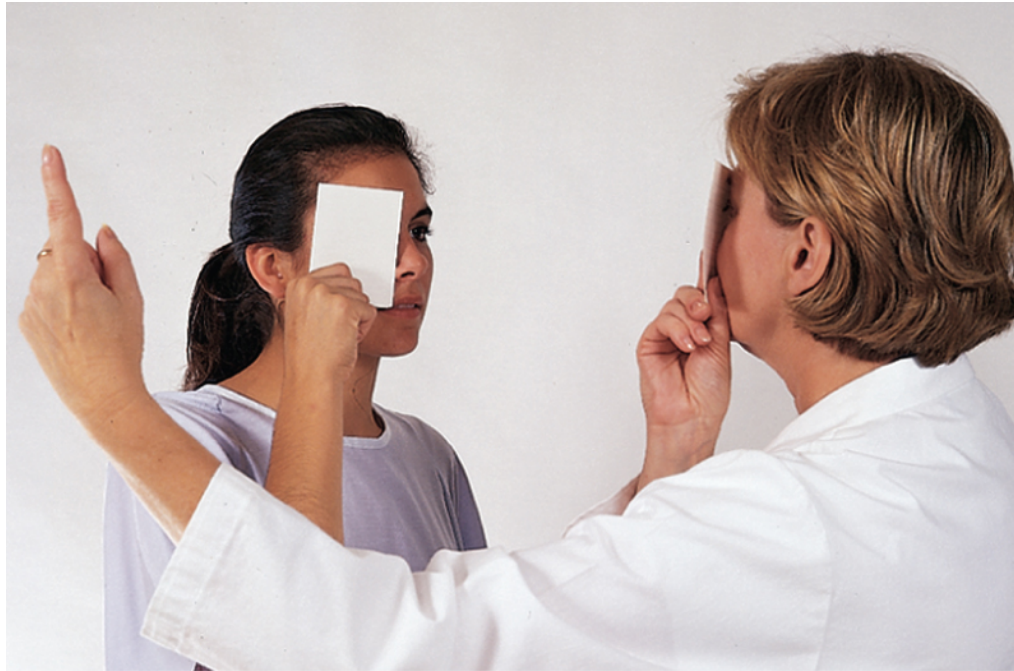
Diagnostic Procedures (Slide 2 of 3)

Visual acuity test: assesses clarity of vision



Diagnostic Procedures (Slide 3 of 3)

Visual field test: measures visual fields when eyes are fixed forward

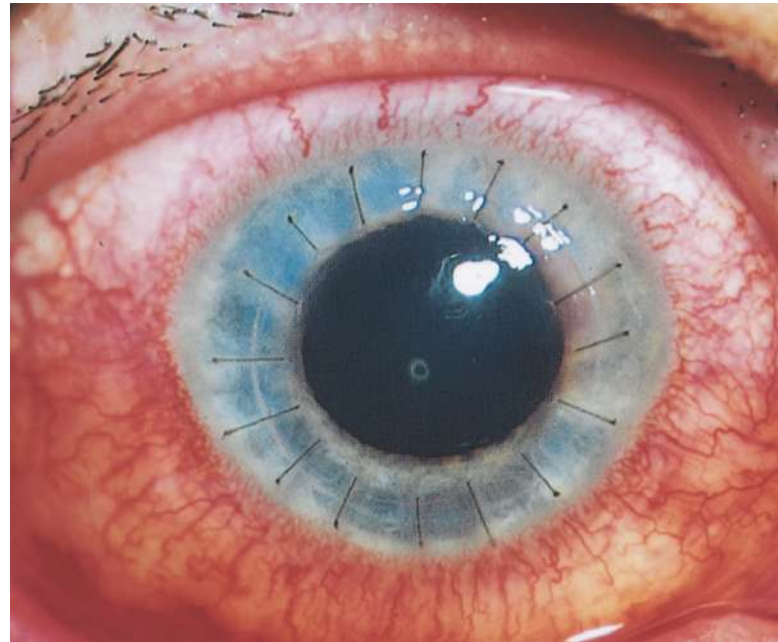


Treatment (Slide 1 of 4)

- **Enucleation:** removal of the entire eyeball
- **Laser photocoagulation:** an argon laser creates an inflammatory reaction that seals retinal tears and leaky blood vessels
- **LASIK:** laser to correct errors of refraction by sculpting the cornea

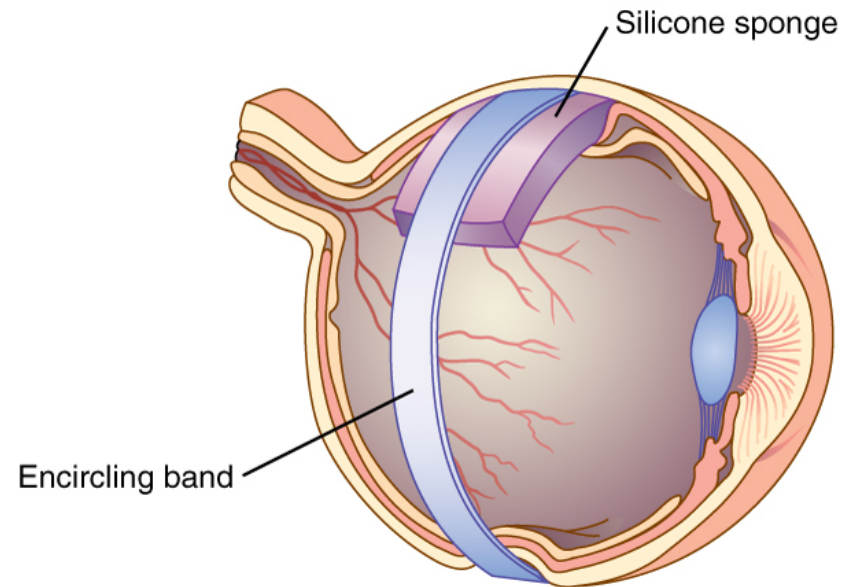
Treatment (Slide 2 of 4)

Keratoplasty: surgical repair of the cornea



Treatment (Slide 3 of 4)

Scleral buckle: suture of a silicone band to the sclera over the detached portion of retina



Treatment (Slide 4 of 4)

- **Phacoemulsification:** ultrasound to break up lens for aspiration for cataract removal
- **Vitrectomy:** removal of vitreous and replacement with a clear solution

Abbreviations – The Eye (Slide 1 of 2)

Abbreviation	Meaning
ALT	Argon laser trabeculoplasty
AMD	Age-related macular degeneration
HEENT	Head, eyes, ears, nose, and throat
IOL	Intraocular lens
IOP	Intraocular pressure
LASER	light a mplification by s timulated e mission of r adiation
LASIK	Laser in situ keratomileusis
OD	Right eye; doctor or optometry
OS	Left eye
OU	Both eyes

Abbreviations – The Eye (Slide 2 of 2)

Abbreviation	Meaning
PERRLA	Pupils equal, round, reactive to light and accommodation
POAG	Primary, open-angle glaucoma
PRK	Photorefractive keratectomy
SLT	Selective laser trabeculoplasty
VA	Visual acuity
VF	Visual field

Chapter 17

Lesson 17.3

The Ear

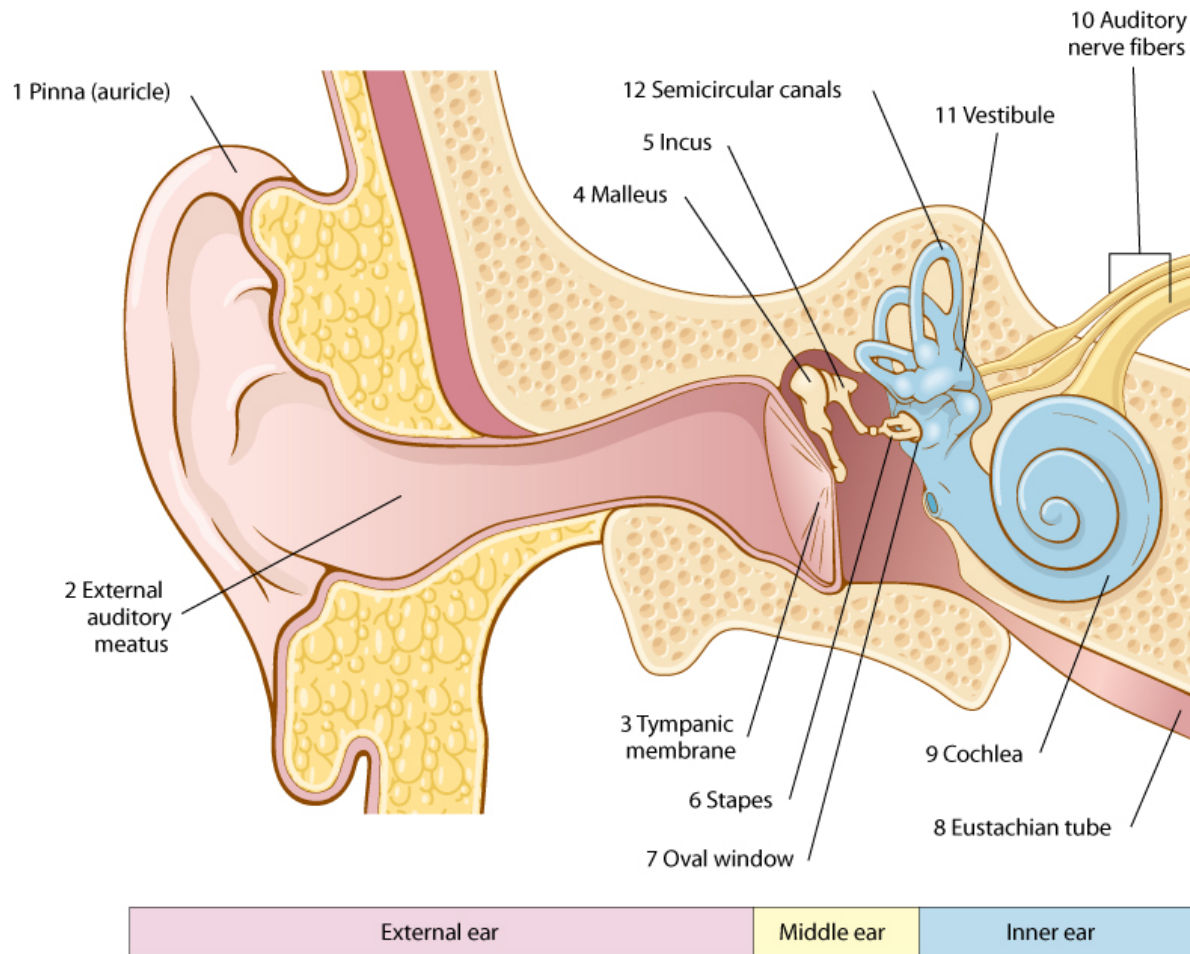
ANATOMY AND PHYSIOLOGY

- The *outer ear* receives sound waves.
- Sound waves travel to the *middle ear*.
- Sound vibrations reach the *inner ear* (labyrinth).

Anatomy of the Ear (Slide 1 of 2)

- Pinna (auricle)
- External auditory meatus (auditory canal)
- Tympanic membrane (eardrum)
- Malleus
- Incus
- Stapes
- Oval window
- Eustachian tube
- Cochlea
- Auditory nerve fibers
- Vestibule
- Semicircular canals

Anatomy of the Ear (Slide 2 of 2)



Ear Anatomy and Physiology

(Slide 1 of 4)

- **Outer ear**

- Pinna or auricle: projecting flap
- External auditory meatus (auditory canal)

Ear Anatomy and Physiology

(Slide 2 of 4)

- **Middle ear**
 - Tympanic membrane (eardrum)
 - Malleus
 - Incus
 - Stapes
 - Oval window
 - Eustachian tube

Ear Anatomy and Physiology

(Slide 3 of 4)

- **Inner ear: labyrinth**
 - Cochlea
 - Auditory nerve fibers

Ear Anatomy and Physiology

(Slide 4 of 4)

- **Balance and equilibrium**
 - Vestibule
 - Semicircular canals

Vocabulary – The Ear (Slide 1 of 14)

Term
auditory canal
auditory meatus
auditory nerve fibers

Vocabulary – The Ear (Slide 2 of 14)

Term	Meaning/Definition
auditory canal	Channel that leads from the pinna to the eardrum
auditory meatus	Auditory canal
auditory nerve fibers	Carry impulses from the inner ear to the brain

Vocabulary – The Ear (Slide 3 of 14)

Term
auditory tube
auricle
cerumen

Vocabulary – The Ear (Slide 4 of 14)

Term	Meaning/Definition
auditory tube	Channel between the middle ear and nasopharynx; eustachian tube
auricle	Flap of the ear; protruding part of external ear, or pinna
cerumen	Waxy substance secreted by the external ear; also called earwax

Vocabulary – The Ear (Slide 5 of 14)

Term
cochlea
endolymph
eustachian tube

Vocabulary – The Ear (Slide 6 of 14)

Term	Meaning/Definition
cochlea	Snail-shaped, spirally wound tube in the inner ear
endolymph	Fluid within the labyrinth of the inner ear
eustachian tube	Auditory tube

Vocabulary – The Ear (Slide 7 of 14)

Term
incus
labyrinth
malleus

Vocabulary – The Ear (Slide 8 of 14)

Term	Meaning/Definition
incus	Second ossicle of the middle ear
labyrinth	Maze-like series of canals of the inner ear
malleus	First ossicle of the middle ear

Vocabulary – The Ear (Slide 9 of 14)

Term
organ of Corti
ossicle
oval window

Vocabulary – The Ear (Slide 10 of 14)

Term	Meaning/Definition
organ of Corti	Sensitive auditory receptor area found in the cochlea of the inner ear
ossicle	Small bone of the ear
oval window	Membrane between the middle ear and the inner ear

Vocabulary – The Ear (Slide 11 of 14)

Term
perilymph
pinna
semicircular canals

Vocabulary – The Ear (Slide 12 of 14)

Term	Meaning/Definition
perilymph	Fluid contained in the labyrinth of the inner ear
pinna	Auricle; flap of the ear
semicircular canals	Passages in the inner ear associated with maintaining equilibrium

Vocabulary – The Ear (Slide 13 of 14)

Term
stapes
tympanic membrane
vestibule

Vocabulary – The Ear (Slide 14 of 14)

Term	Meaning/Definition
stapes	Third ossicle of the middle ear
tympanic membrane	Membrane between the outer and the middle ear; eardrum
vestibule	Central cavity of the labyrinth, connecting the semicircular canals and the cochlea

QUICK QUIZ (Slide 3 of 3)

- 3. What is the snail-shaped, spirally wound tube in the inner ear that contains hearing-sensitive receptor cells?**
- A. Pinna
 - B. Auricle
 - C. Ossicle
 - D. Cochlea

Combining Forms Ear Terminology

(Slide 1 of 2)

Combining Form	Meaning
acous/o	hearing
audi/o	hearing
audit/o	hearing
aur/o	ear
auricul/o	ear
cochle/o	cochlea
mastoid/o	mastoid process

Combining Forms Ear – Terminology

(Slide 2 of 2)

Combining Form	Meaning
myring/o	eardrum, tympanic membrane
ossicul/o	ossicle
ot/o	ear
salping/o	Eustachian tube
staped/o	stapes
tyman/o	eardrum, tympanic membrane
vestibul/o	vestibule

Terminology Ear- Suffixes (Slide 1 of 1)

Suffix	Meaning
-acusis or-cusis	hearing
-meter	instrument for measure
-otia	ear condition

Symptoms and Pathologic Conditions

(Slide 1 of 3)

- **Acoustic neuroma:** benign tumor arising from 8th cranial nerve, causes tinnitus, vertigo, dizziness, and decreased hearing
- **Cholesteatoma:** skin cells and cholesterol in a sac in the middle ear (cyst-like mass associated with chronic infections)
- **Deafness:** loss of ability to hear

Symptoms and Pathologic Conditions

(Slide 2 of 3)

- **Ménière disease:** disorder of the labyrinth with elevated endolymph pressure in the cochlea and semicircular canals causing tinnitus, sensitivity to sound, progressive hearing loss, headache, nausea, and vertigo
- **Otitis media:** inflammation of the middle ear
- **Otosclerosis:** hardening of bony tissue in the labyrinth

Symptoms and Pathologic Conditions

(Slide 3 of 3)

- **Tinnitus:** sensation of noises (ringing, buzzing, whistling, booming) in the ears
- **Vertigo:** sensation of irregular motion (whirling) from disease of the inner ear or nerve carrying messages from the semicircular canals

Clinical Procedures (Slide 1 of 2)

- **Audiometry:** an audiometer is an electric device used to determine hearing loss by frequency
- **Cochlear implant:** a surgically implanted device that allows sensorineural hearing-impaired persons to understand speech
- **Ear thermometry:** body temperature measured with infrared radiation from the eardrum

Clinical Procedures (Slide 2 of 2)

- **Otoscopy:** visual examination of the ear with a small, hand-held scope
- **Tuning fork test:** hearing tests using a vibrating fork
 - **Rinne:** fork is place against the mastoid bone (bone conduction) and in front of the auditory meatus (air conduction)
 - **Weber:** fork is placed on the center of the forehead; normal hearing has equal loudness in both ears

Abbreviations – The Ear

Abbreviation	Meaning
AD	Right ear
AOM	Acute otitis media
AS	Left ear
EENT	Eyes, ears, nose, and throat
ENG	Electronystagmography
ENT	Ears, nose, and throat
ETD	Eustachian tube dysfunction
HEENT	Head, eyes, ears, nose, and throat
PE tube	Pressure-equalizing tube
SOM	Serous otitis media

Review Sheet – Combining Forms

(Slide 1 of 14)

Combining Form
acous/o
ambly/o
aque/o
audi/o
audit/o
aur/o

Review Sheet – Combining Forms

(Slide 2 of 14)

Combining Form	Meaning
acous/o	hearing
ambly/o	dull, dim
aque/o	water
audi/o	hearing
audit/o	hearing
aur/o	ear

Review Sheet – Combining Forms

(Slide 3 of 14)

Combining Form
auricul/o
blephar/o
conjunctiv/o
cor/o
corne/o
cycl/o

Review Sheet – Combining Forms

(Slide 4 of 14)

Combining Form	Meaning
auricul/o	ear
blephar/o	eyelid
conjunctiv/o	conjunctiva
cor/o	pupil
corne/o	cornea
cycl/o	ciliary body

Review Sheet – Combining Forms

(Slide 5 of 14)

Combining Form
dacry/o
dipl/o
glauc/o
ir/o
irid/o
kerat/o
lacrim/o

Review Sheet – Combining Forms

(Slide 6 of 14)

Combining Form	Meaning
dacry/o	tears
dipl/o	double
glauc/o	gray
ir/o	iris
irid/o	iris
kerat/o	cornea
lacrim/o	tears

Review Sheet – Combining Forms

(Slide 7 of 14)

Combining Form
mastoid/o
mi/o
mydr/o
myring/o
nyct/o
ocul/o

Review Sheet – Combining Forms

(Slide 8 of 14)

Combining Form	Meaning
mastoid/o	mastoid process
mi/o	smaller, less
mydr/o	widen, enlarge
myring/o	eardrum
nyct/o	night
ocul/o	eye

Review Sheet – Combining Forms

(Slide 9 of 14)

Combining Form
ophthalm/o
opt/o
optic/o
ossicul/o
palpebr/o
papill/o
phac/o

Review Sheet – Combining Forms

(Slide 10 of 14)

Combining Form	Meaning
ophthalm/o	eye
opt/o	eye
optic/o	eye
ossicul/o	ossicle
palpebr/o	eyelid
papill/o	optic disc
phac/o	lens of the eye

Review Sheet – Combining Forms

(Slide 11 of 14)

Combining Form
phak/o
phot/o
presby/o
pupill/o
retin/o
salpin/o
scler/o

Review Sheet – Combining Forms

(Slide 12 of 14)

Combining Form	Meaning
phak/o	lens of the eye
phot/o	light
presby/o	old age
pupill/o	pupil
retin/o	retina
salpin/o	Eustachian tube
scler/o	sclera (white of eye)

Review Sheet – Combining Forms

(Slide 13 of 14)

Combining Form
scot/o
staped/o
tympan/o
uve/o
vestibul/o
vitre/o
xer/o

Review Sheet – Combining Forms

(Slide 14 of 14)

Combining Form	Meaning
scot/o	darkness
staped/o	stapes
tympan/o	eardrum
uve/o	uvea
vestibul/o	vestibule
vitre/o	glassy
xer/o	dry

Review Sheet – Suffixes (Slide 1 of 4)

Suffix
-acusis
-cusis
-meter
-metry
-opia

Review Sheet – Suffixes (Slide 2 of 4)

Suffix	Meaning
-acusis	hearing
-cusis	hearing
-meter	instrument for measure
-metry	process of measurement
-opia	vision

Review Sheet – Suffixes (Slide 3 of 4)

Suffix
-opsia
-otia
-phobia
-plegic
-tropia

Review Sheet – Suffixes (Slide 4 of 4)

Suffix	Meaning
-opsia	vision
-otia	ear condition
-phobia	fear
-plegic	paralysis; palsy
-tropia	to turn